House Prices: Machine Learning Capstone Project

Regression, Classification, and Unsupervised Clustering

Dataset: House Prices - Advanced Regression Techniques (Kaggle)

Dataset size: 1,460 rows and 81 columns in the original training file

This project applies three machine learning approaches to the Kaggle House Prices dataset: Linear Regression to estimate house sale prices, Logistic Regression to classify houses into two price groups, and K-Means to explore groups in the feature data without using target labels.

Project notebook: Open Google Colab notebook

GitHub repository: ML-Capstone-Project

Dataset source: Kaggle competition page

1. Introduction

The objective of this project is to demonstrate a complete introductory machine learning workflow using residential property data. The dataset contains numerical and categorical attributes describing homes, along with the SalePrice target. The work includes data inspection, preprocessing, model training, evaluation, and visualization.

2. Exploratory Data Analysis and Preprocessing

The original training dataset contains 1,460 observations and 81 columns. It includes 35 integer columns, 3 floating-point columns, and 43 categorical/object columns. Missing values were found in 19 columns. No duplicate rows were detected. Boxplots of SalePrice and GrLivArea showed potential outliers; these were retained because unusually priced or sized houses may be valid observations.

The Id column was removed from the predictors, and SalePrice was kept as the regression target. The data was split into training and testing sets using an 80/20 split (random_state=42), resulting in 1,168 training rows and 292 testing rows.

Preprocessing was fitted using the training data only to reduce data leakage. For numerical features, missing values were imputed with the median and values were standardized using StandardScaler. For categorical features, missing values were imputed using the most frequent category and categories were transformed using OneHotEncoder(handle_unknown='ignore'). For fields where a missing value indicates that a feature is absent (such as a garage, basement, pool, or fence), a 'None' category was used. The processed training matrix had 300 columns; the testing matrix had the same 300 columns.

3. Regression: Linear Regression

Linear Regression was trained to predict the continuous SalePrice value. The model was evaluated on the held-out test set using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared.

Metric	Test result
MSE	4,267,350,801.06
RMSE	65,324.96
R-squared (R^2)	0.4437

The R^2 score indicates that the model explains approximately 44.37% of the variation in house prices in the test set. The RMSE indicates a typical prediction error magnitude of about 65,325 price units. These results suggest that this baseline linear model has room for improvement.

4. Classification: Logistic Regression

The original dataset does not contain a classification label. Therefore, a binary target named PriceCategory was derived from SalePrice using the median SalePrice of the training set (165,000). A house was assigned class 0 when its price was less than or equal to 165,000, and class 1 when its price was greater than 165,000. The same training-derived threshold was applied to the test set.

Logistic Regression was trained to predict the two price categories. The test-set results were:

Metric	Test result
Accuracy	94.86%
Precision (class 1)	91.85%
Recall (class 1)	96.88%

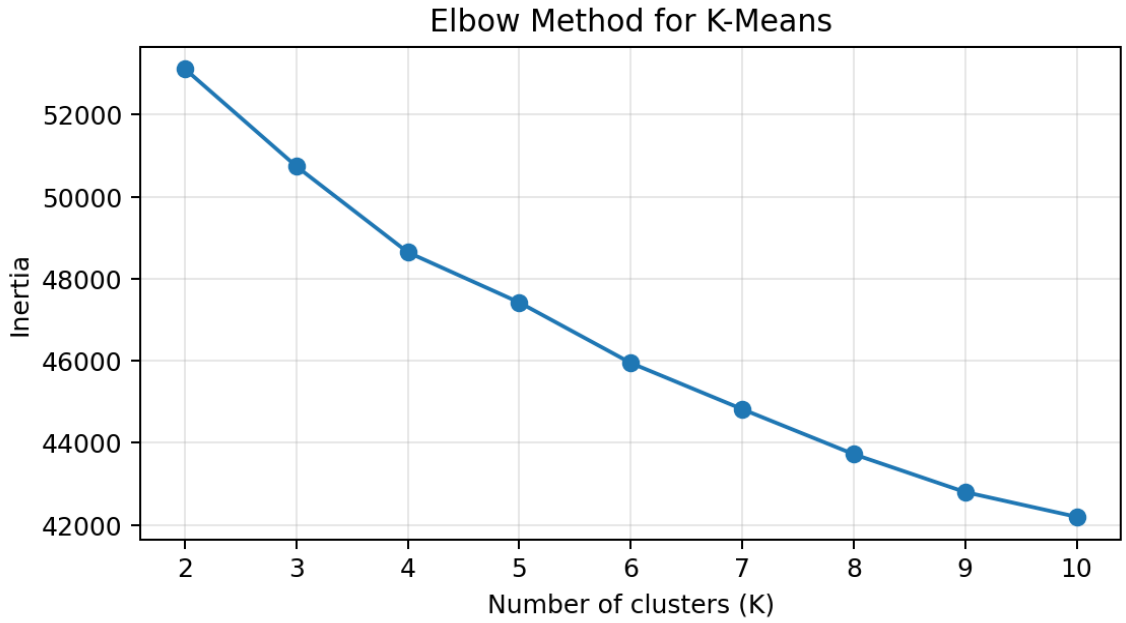
Confusion matrix (rows = actual class; columns = predicted class):

	Predicted 0	Predicted 1
Actual 0	153	11
Actual 1	4	124

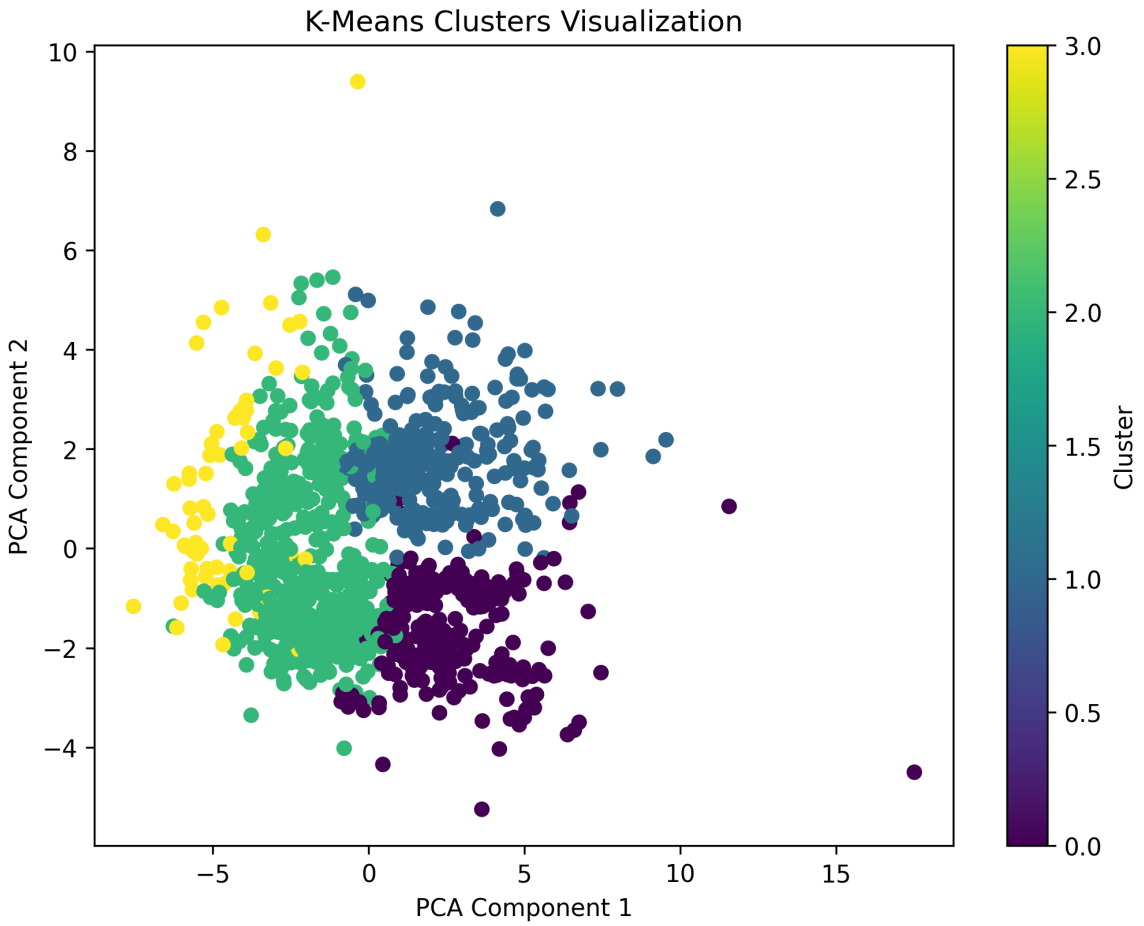
The classifier correctly classified 277 out of 292 test examples. The derived target makes the task a binary high-price versus not-high-price classification problem; it is not an original label supplied by Kaggle.

5. Clustering: K-Means

K-Means was applied to the processed training features without using SalePrice or the derived classification target. The Elbow Method was examined for values of K from 2 to 10. Based on the observed curve, K=4 was selected as an approximate choice for the final clustering model.



The final K-Means model used four clusters. Its Silhouette Score was 0.1044. This relatively low score suggests that the clusters are not strongly separated and that some overlap exists between groups. The PCA chart below projects the high-dimensional processed data into two dimensions for visual inspection; the projection is a visualization and does not itself define the clusters.



6. Conclusion

This project completed three machine learning tasks using the House Prices dataset. Linear Regression provided a baseline for predicting sale prices, while Logistic Regression achieved 94.86% accuracy on the derived binary price-category task. K-Means identified four exploratory groups, although the low Silhouette Score indicates limited separation. Overall, the project demonstrates data inspection, missing-value handling, categorical encoding, scaling, train/test splitting, model evaluation, and visualization. Future improvements could include testing more advanced regression models, tuning hyperparameters, and exploring alternative clustering methods or feature representations.

7. Project Resources

Google Colab notebook: Open notebook
GitHub repository: Open repository
Dataset: Kaggle House Prices

Note: Before final submission, make sure the GitHub repository contains the notebook, results/ visualizations and CSV files, README.md, and this report saved as project_report.pdf. Also verify that the Colab sharing permission is set to anyone with the link as Viewer.